

Login to ICE (with VPN)

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>

Interactive Apps

Applications

ANSYS Workbench

Jupyter

MATLAB

Ollama + Jupyter (Beta)

RStudio

vLLM + Jupyter

Desktops

Interactive Desktop

IDEs

Coder

Microsoft VS Code

Tutorials

End-to-End-LLM Bootcamp

Makerspace Jupyter Tutorial

Coder

This app will launch [Coder](#) on a compute node. Coder is an IDE based on the open-source components of Microsoft VS Code.

This OOD application will no longer receive updates. Please switch to the [Microsoft VS Code OOD Application](#), which provides the official version of Microsoft VS Code.

Modules

Anaconda 3 - 2023.03

PACE-Apps modules to load before starting VS Code.

- VS Code extensions will only find PACE-apps if modules are specified here!
- To run custom module loads and shell commands, specify "Custom Modules" and fill in the text field below.

Custom Pre-load Commands

Optional. Bash commands to run before starting VS Code. Example: `module load cuda/11.7`

- You can enter multiple lines and arbitrary shell commands.
- Only used when "Custom Pre-load" is selected in the "Modules" dropdown, above

Quality of Service

Default (none)

Node Type

Intel CPU

Cores

6

Number cores (or "CPUs") for session. VS Code is more responsive with ≥ 4 cores.

Connect to Coder

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>

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-  MATLAB
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-  RStudio
-  vLLM + Jupyter

Desktops

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-  Coder
-  Microsoft VS Code

Tutorials

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-  Makerspace Jupyter Tutorial

Coder (5583880)

1 node | 6 cores | Running

Host: `>_atl1-1-01-005-1-1.pace.gatech.edu`

Delete

Created at: 2026-08-19 23:16:21 EDT

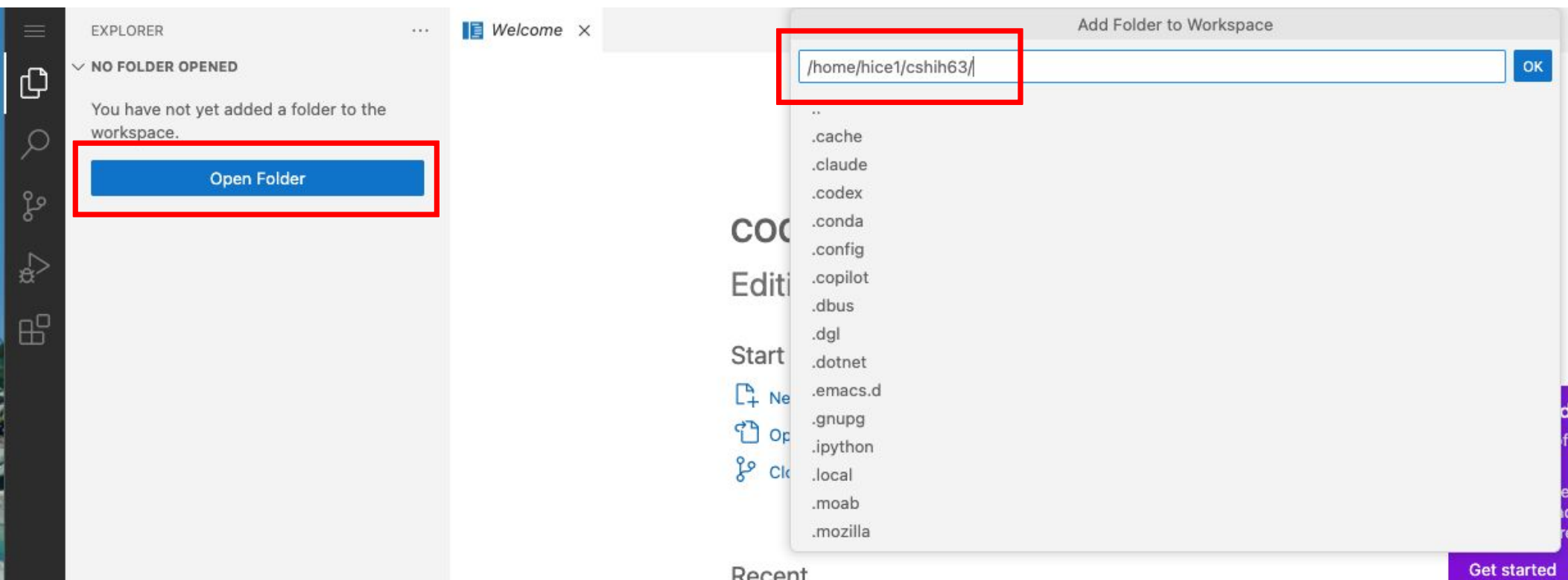
Time Remaining: 3 hours and 58 minutes

Session ID: a8a9baae-10ec-42f4-b736-3cc26f17d07b

Connect

Open Home Directory

- <https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard/>



New Terminals

The screenshot shows the Visual Studio Code (VS Code) interface. On the left, the 'Terminal' menu is open, with 'New Terminal' highlighted. The menu also shows 'Split Terminal', 'Run Task...', 'Run Build Task...', 'Run Active File', 'Run Selected Text', 'Show Running Tasks...', 'Restart Running Task...', 'Terminate Task...', 'Configure Tasks...', and 'Configure Default Build Task...'. The background shows the 'code-server' welcome screen with the text 'code-server Editing evolved'. Below this, there are sections for 'Start' (New File..., Open File..., Clone Git Repository...) and 'Recent' (job (Workspace) ~/ondemand/data/sys/dashboard/batch_connect/sys/bc_c...).

Terminal Menu:

- File
- Edit
- Selection
- View
- Go
- Run
- Terminal**
- Help
- Sign out of code-server
- > .dotnet
- > .emacs.d
- > .gnupg
- > .ipython
- > .local
- > .moab
- > .mozilla
- > .nv
- > .ssh
- > .triton
- > .vnc

New Terminal Sub-menu:

- New Terminal
- Split Terminal
- Run Task...
- Run Build Task...
- Run Active File
- Run Selected Text
- Show Running Tasks...
- Restart Running Task...
- Terminate Task...
- Configure Tasks...
- Configure Default Build Task...

code-server
Editing evolved

Start

- New File...
- Open File...
- Clone Git Repository...

Recent

- job (Workspace) ~/ondemand/data/sys/dashboard/batch_connect/sys/bc_c...

Check Binaries

- `source /storage/ice-shared/cs3220/labs_setup.sh`
- With **iverilog** installed, you are ready to go

```
~$ source /storage/ice-shared/cs3220/labs_setup.sh
Lmod has detected the following error: These module(s) or extension(s) exist but cannot be loaded
as requested: "gtkwave"
Try: "module spider gtwave" to see how to load the module(s).
```

```
~$ iverilog
iverilog: no source files.
```

```
Usage: iverilog [-EiSuvV] [-B base] [-c cmdfile|-f cmdfile]
               [-g1995|-g2001|-g2005|-g2005-sv|-g2009|-g2012] [-g<feature>]
               [-D macro[=defn]] [-I includedir] [-L moduledir]
               [-M [mode=]depfile] [-m module]
               [-N file] [-o filename] [-p flag=value]
               [-s topmodule] [-t target] [-T min|typ|max]
               [-W class] [-y dir] [-Y suf] [-l file] source_file(s)
```

See the man page for details.

```
~$ |
```

Install Extensions - iverilog/gtkwave

EXTENSIONS: MARKETPLACE

gkwave



Icarus Verilog + GTKWave Runn...

Run Verilog/SystemVerilog testbe...

j054n



dglog

Run Verilog/SystemVerilog testbe...

j054n

1K

Install

Extension: Icarus Verilog + GTKWave Runner for VSCode



Icarus Verilog + GTKWave Runner for VSCode

j054n | 3,384 | ★★★★★

Run Verilog/SystemVerilog testbench with GTKWave support.

Disable | Uninstall | ⚙️

This extension is enabled globally.

DETAILS

FEATURE CONTRIBUTIONS

EXTENSION PACK

that **iverilog** and **gkwave** have been added to your environment PATH variable. If you need help, check out the installation guide for [Windows](#) and [other OSes](#).

Usage

The extension adds in VSCode:

- two buttons in the titlebar of any **Verilog/SystemVerilog** file:
 - Run testbench simulation
 - Stop simulation and clean build files
- a **status** item positioned in the lower-right corner
- **console log output** of commands and running testbench

Categories

Programming Languages

Extension Resources

[Marketplace](#)

[Repository](#)

[License](#)

j054n

More Info

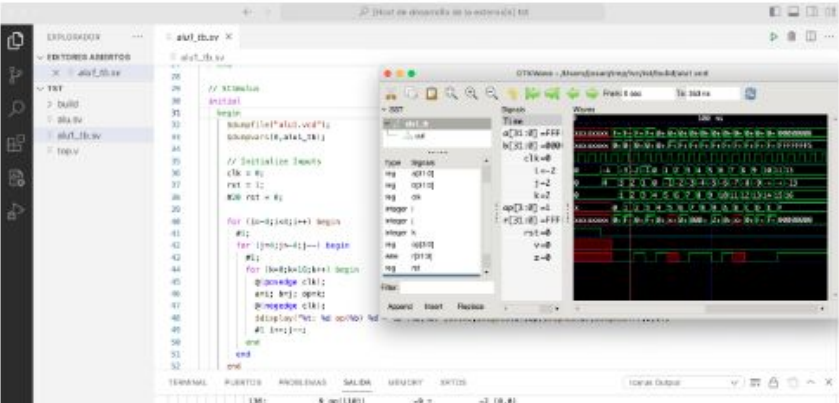
Published1/26/2024,

15:23:36

Last1/26/2024,

released 15:23:36

Identifier dglab.dglog



Install Extensions - SystemVerilog

EXTENSIONS: MARKETPLACE

systemverilog

- Vide - Verilog/SystemVerilog** 620
A Modern SystemVerilog Coding I...
pascal-lab [Install](#)
- Vide - Verilog/System...** 6K ★ 5
A Modern SystemVerilog Coding I...
pascal-lab [Install](#)
- SystemVerilog Pro** 4K ★ 5
Fast SystemVerilog, Verilog, and ...
Taher Anaya [Install](#)
- Verilog-HDL/Syste...** 304K ★ 5
Verilog-HDL/SystemVerilog suppo...
mshr-h [Install](#)
- SystemVerilog Tools** 1K
Verilog/SystemVerilog formatter, ...
JayceVane [Install](#)
- slang-server: Verilog/...** 6K ★ 5
Verilog and SystemVerilog suppor...
Hudson-River-Trading [Install](#)
- SystemVerilog LSP - Synt...** 1K
Simple SystemVerilog language s...
mashurr [Install](#)
- VeriGood - SystemVerilo...** 3K
Verilog/SystemVerilog formatter w...
FabioOliveira [Install](#)
- SystemVerilog and Verilo...** 5K
Beautify SystemVerilog and Verilo...
bmpenuelas [Install](#)
- HDL BlockSeal** 330
Auto-inserts closing keywords (en...
VENOMNBB [Install](#)
- DigitalJS Schematic Vie...** 332
Synthesises Verilog/SystemVerilo...
VENOMNBB [Install](#)

Extension: Verilog-HDL/SystemVerilog x test.v

Verilog-HDL/SystemVerilog v1.29.0

mshr-h | 304,024 | ★★★★★ (1)

Verilog-HDL/SystemVerilog support for VS Code

[Install in ondemand-ice.pace.gatech.edu](#) ⚙️

[DETAILS](#) [FEATURE CONTRIBUTIONS](#) [CHANGELOG](#)

HDL support for VS Code

HDL support for VS Code with syntax highlighting, snippets, classic Ctags-backed editing features, linting, formatting, waveform viewing, inactive preprocessor highlighting, and optional external language servers.

openssf best practices in progress 16% [Ask DeepWiki](#)

Categories

- Programming Languages
- Snippets
- Linters
- Formatters
- Visualization

Extension Resources

- [Marketplace](#)
- [Repository](#)
- [License](#)
- [mshr-h](#)

More Info

Published 11/10/2021, 06:04:29

Last released 7/11/2026, 01:34:20

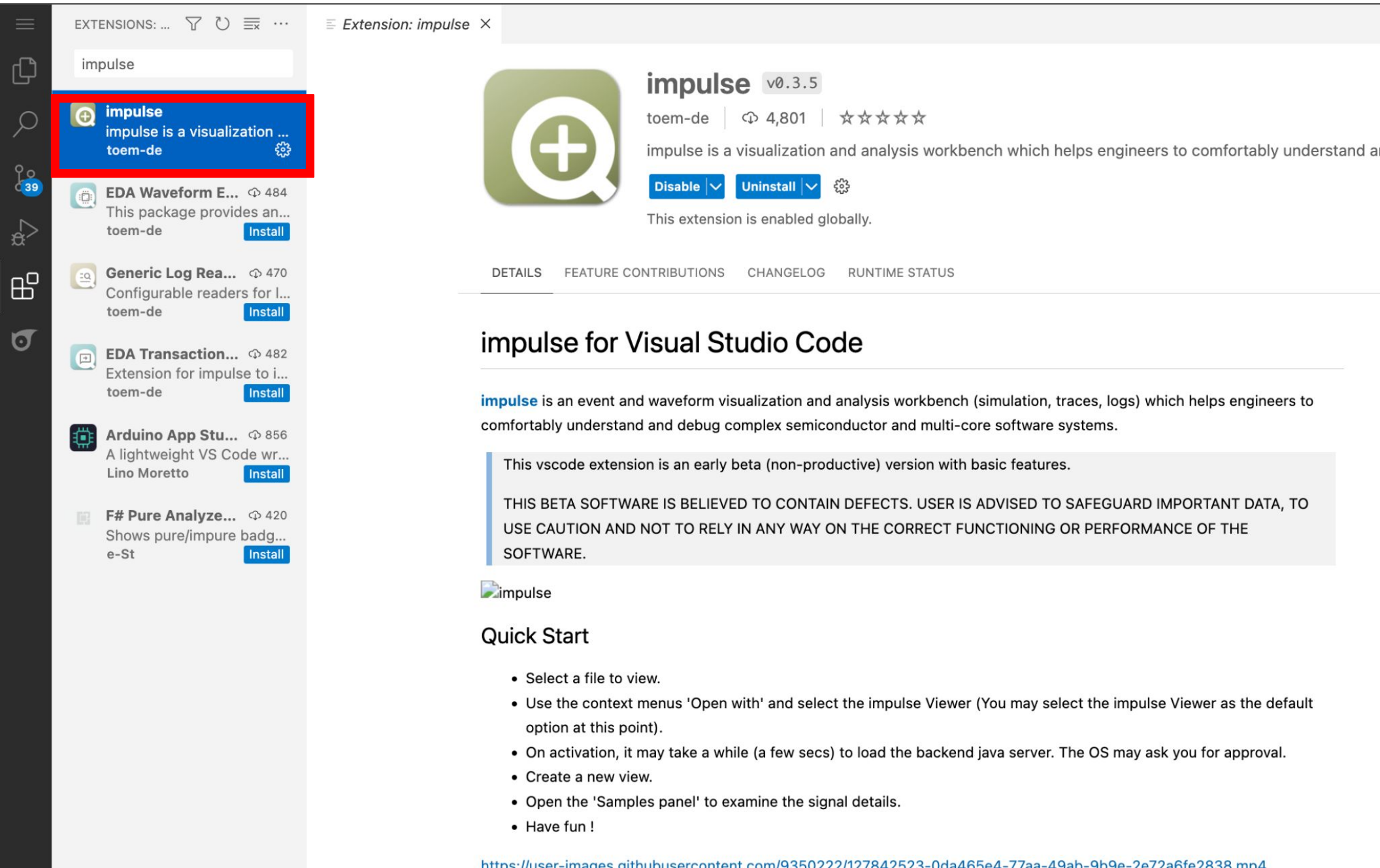
Identifier mshr-h.veriloghdl

```
1 module some_mod(  
2     input wire clk,  
3     input wire rst,  
4     input wire in,  
5     output reg out  
6 );  
7  
8     reg rand_reg;  
9  
10    always@* begin  
11        out = ~in  
12    end  
13  
14 endmodule
```

Installation

Install the extension from the [VS Code Marketplace](#) or [Open VSX Registry](#)

Install Extensions - SystemVerilog



The screenshot shows the Visual Studio Code interface with the Extensions view open. In the left sidebar, the 'impulse' extension by 'toem-de' is highlighted with a red box. The main panel displays the details for the 'impulse' extension, version 0.3.5. The extension is described as a visualization and analysis workbench for engineers. It has 4,801 downloads and a 5-star rating. The extension is currently installed and enabled globally. Below the extension details, there is a section titled 'impulse for Visual Studio Code' which provides a description of the extension and a 'Quick Start' guide.

EXTENSIONS: ... **Extension: impulse** X

impulse v0.3.5
toem-de | 4,801 | ★★★★★
impulse is a visualization and analysis workbench which helps engineers to comfortably understand and debug complex semiconductor and multi-core software systems.

Disable **Uninstall** 

This extension is enabled globally.

DETAILS FEATURE CONTRIBUTIONS CHANGELOG RUNTIME STATUS

impulse for Visual Studio Code

impulse is an event and waveform visualization and analysis workbench (simulation, traces, logs) which helps engineers to comfortably understand and debug complex semiconductor and multi-core software systems.

This vscode extension is an early beta (non-productive) version with basic features.

THIS BETA SOFTWARE IS BELIEVED TO CONTAIN DEFECTS. USER IS ADVISED TO SAFEGUARD IMPORTANT DATA, TO USE CAUTION AND NOT TO RELY IN ANY WAY ON THE CORRECT FUNCTIONING OR PERFORMANCE OF THE SOFTWARE.

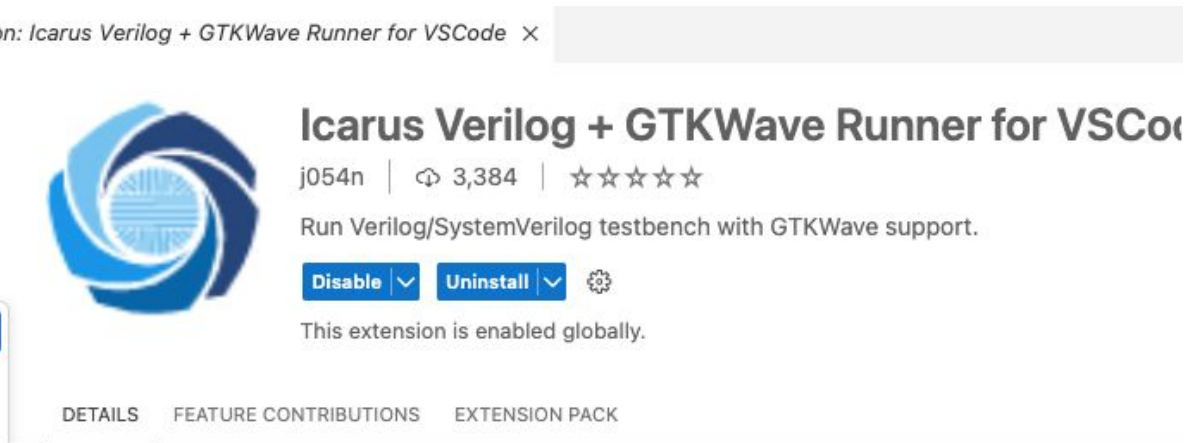
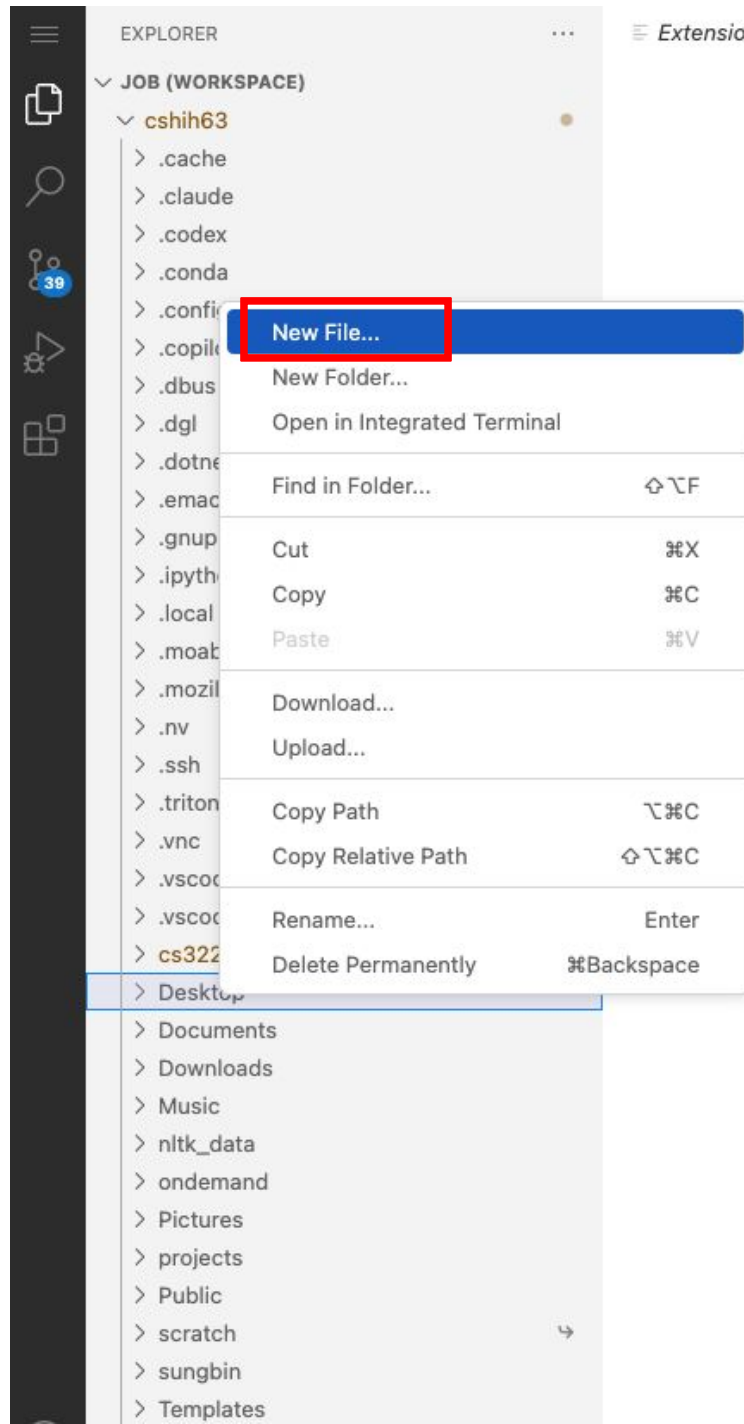
 Impulse

Quick Start

- Select a file to view.
- Use the context menus 'Open with' and select the impulse Viewer (You may select the impulse Viewer as the default option at this point).
- On activation, it may take a while (a few secs) to load the backend java server. The OS may ask you for approval.
- Create a new view.
- Open the 'Samples panel' to examine the signal details.
- Have fun !

<https://user-images.githubusercontent.com/9350222/127842523-0da465e4-77aa-49ab-9b9e-2e72a6fe2838.mp4>

New File



that **iverilog** and **gtkwave** have been added to your environment PATH variable. If you need help, check out the installation guide for [Windows](#) and [other OSes](#).

Usage

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 - Run testbench simulation
 - Stop simulation and clean build files
- a **status** item positioned in the lower-right corner
- **console log output** of commands and running testbench

Categories

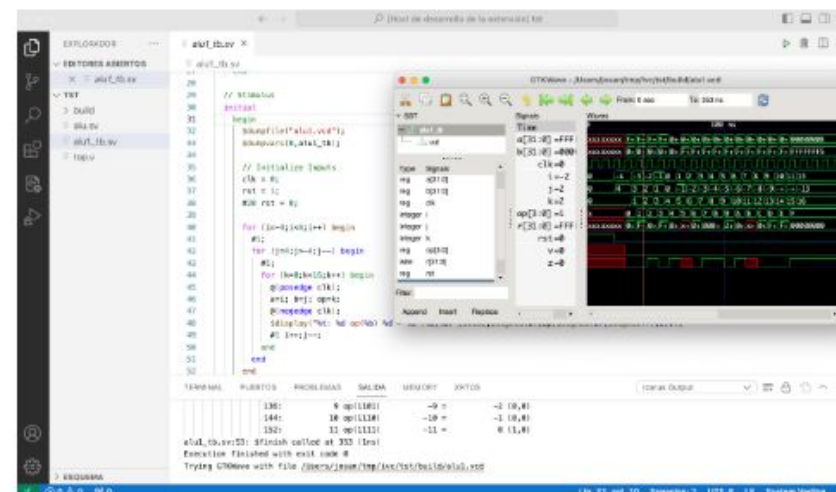
Programming
Languages

Extension Resources

[Marketplace](#)
[Repository](#)
[License](#)
[j054n](#)

More Info

Published 1/26/2024,
15:23:36
Last 1/26/2024,
released 15:23:36
Identifier dglab.dglog



Test Adder

```
1 module testbench;
2   reg [3:0] a;
3   reg [3:0] b;
4   wire [4:0] sum;
5
6   adder dut (
7     .a(a),
8     .b(b),
9     .sum(sum)
10  );
11
12  initial begin
13    $dumpfile("adder.vcd");
14    $dumpvars(0, testbench);
15  end
16
17  integer i;
18  integer errors;
19  initial begin
20    errors = 0;
21    $display("a b sum expected");
22    $display("-----");
23    for (i = 0; i < 8; i = i + 1) begin
24      a = i;
25      b = i + 1;
26      #10;
27      $display("%4d %4d %5d %5d %s",
28        a, b, sum, a + b,
29        (sum == (a + b)) ? "OK" : "FAIL");
30      if (sum != (a + b))
31        errors = errors + 1;
32    end
33    if (errors == 0)
34      $display("PASSED");
35    else
36      $display("FAILED - %0d error(s)", errors);
37    $finish;
38  end
39
40 endmodule
```

testbench.v

```
module adder (
  input [3:0] a,
  input [3:0] b,
  output [4:0] sum
);

  assign sum = a + b;

endmodule
```

test.v

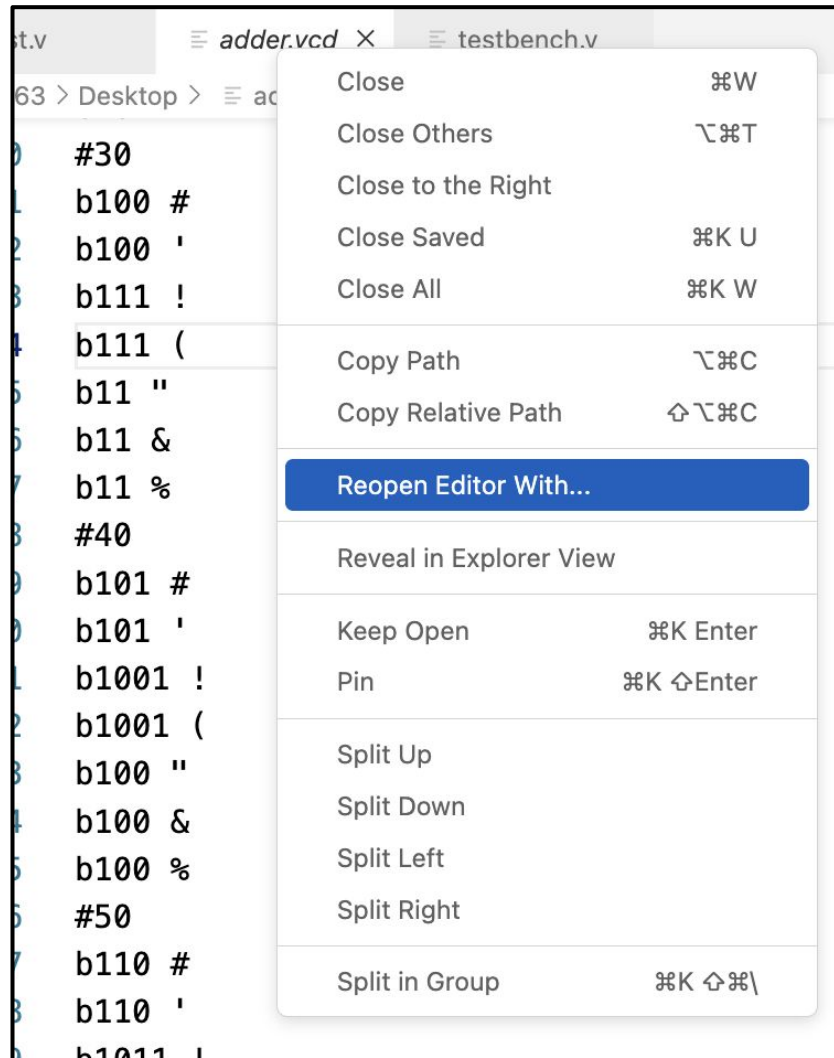
```
~$ cd Desktop/
~/Desktop$ iverilog test.v testbench.v
~/Desktop$ vvp ./a.out
```

a	b	sum	expected	
0	1	1	1	OK
1	2	3	3	OK
2	3	5	5	OK
3	4	7	7	OK
4	5	9	9	OK
5	6	11	11	OK
6	7	13	13	OK
7	8	15	15	OK

PASSED
~/Desktop\$ |

Check waveform

- ❏ Right click vcd file -> Reopen Editor ...
-> Signal Viewer



Check waveform

❏ On the right panel, Add New View

The screenshot displays a digital logic simulation environment. On the left, a 'Value Change Dump' panel shows a hierarchy of testbench components: testbench, dut, sum, a, b, errors, and i. The central area shows a waveform with a time axis from 0 to 50ns. The waveform data is as follows:

Signal	0	10s	20s	30s	40s	50s
x01	x0	x3	x5	x7	x9	x0
x0	x1	x2	x3	x4	x5	x6
x1	x2	x3	x4	x5	x6	x7
x00000000						
x00000000	x00000001	x00000002	x00000003	x00000004	x00000005	x00000006

On the right, the 'View' panel is open, showing the 'add hierarchy' view. The 'Name' is 'add hierarchy' and the 'Description' is 'Create a hierarchy of all signals and scopes'. The 'Axis type' is set to 'Linear' and the 'Domain base' is 'Use signal domain base'. The 'Mode' is 'Use signal domain base'. The 'Domain class' is 'Time'. A red box highlights the 'Create a hierarchy of all signals and scopes' button, and a red arrow points to it from the 'Add a flat list of all signals' button.

Or download the vcd file and use gtkwave

Download Files

